20 FEB 2026

AADK TASK 3

Modifiers, Spacing & Responsive Behavior Report
(hands-on + documented)

MindMatrix

Introduction

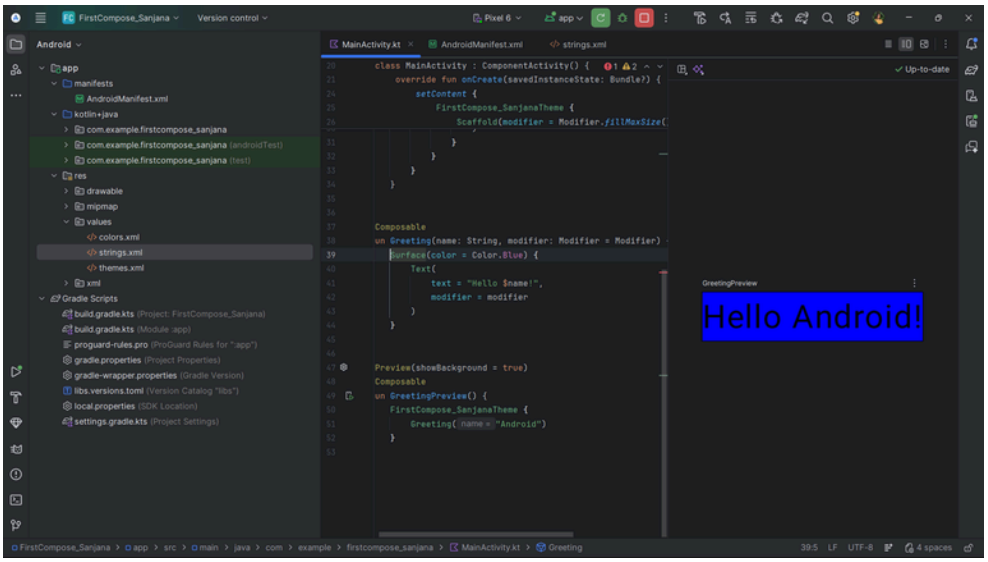
In Jetpack Compose, the Modifier system controls layout size, spacing, alignment, styling, and responsiveness.
Modifiers are chained together, and their order matters, which can significantly affect how UI elements behave across different screen sizes.
This report documents three modifier experiments and analyzes layout behavior on:

- Compose Preview (Desktop view)
- Small phone emulator
- Large phone / tablet emulator

Experiment 1 — padding() vs Background Order

Modifier Tested:
padding() and background()

Padding should create space around the content, while background should color the element's visible area. I expected padding to always add space outside the colored region.



Experiment 2 — fillMaxWidth() + weight()

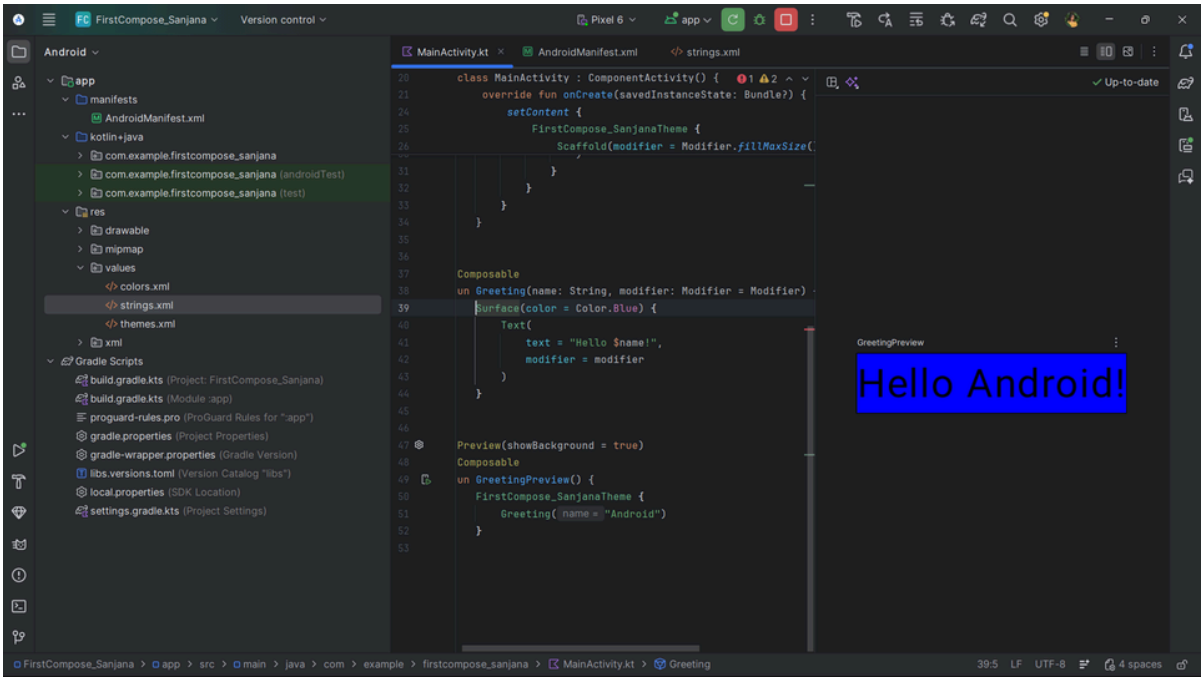
Modifier Tested:
fillMaxWidth() and weight() inside a Row

Using weight() should distribute horizontal space proportionally between elements.
fillMaxWidth() should stretch an element across available width.

Experiment 3 — align() & Arrangement.SpaceBetween

align() inside Column
Arrangement.SpaceBetween in Row

Arrangement.SpaceBetween should push elements to opposite edges.
align() should override default alignment inside parent layout.



Accessibility Mini Check

Observed Issue:
In Experiment 2 (small phone), buttons became crowded with minimal spacing.

Potential Accessibility Problem:

- Small touch targets
- Reduced readability

Suggested Fix:

- Increase minimum button height
- Add consistent padding
- Use Spacer() for breathing room
- Follow minimum 48dp touch target guideline